WHARTON INVESTMENT COMPETITION

THE FAMA FRENCH FRIES

ARNAV DAS, ANEESH CHATRATHI, ANANYA VADERA, ARJUN GUPTA, & AJAY VINJAMURI

Username: Rock Ridge Team Bull-8829977
User ID: 8830015
Advisor(Teacher): Howard Goldberg
Advisor Email: Howard.Goldberg@lcps.org
Team Leader's Name: Arnav Das
Team Leader Email: 901125@lcps.org
School: Rock Ridge High School
Ashburn, VA, United States

Dec. 13, 2024

ADVISOR REFLECTION

Over the past 10 weeks, I have watched this team evolve from five individuals with unique talents into a cohesive team that leverages each member's strengths and inspires one another to reach for excellence. They fostered a team dynamic where everyone felt safe to share their ideas openly, enabling honest and constructive feedback. I took a hands-off approach to this team, giving them guidance when they needed it and serving as a sounding board for their ideas. As an economics teacher, I noticed that earlier in the competition, they were primarily focusing on the quantitative aspect of stock forecasting and weren't placing a strong emphasis on qualitative aspects like macroeconomic trends. After teaching them the basics and urging them to do some further reading, I was pleased with the depth of their macroeconomic analysis in their final strategy. Overall, I thought this team demonstrated a growth mindset and learned a great deal about investing from this competition.

TRADE NOTE ANALYSIS

Vanguard

GE VERNOVA

SONY

ETF

Vanguard Health Care (VHT)

Trade Note

VHT is a fabulous ETF for a long term hold as the healthcare industry is currently undervalued after falling due to the over-optimism during the peak of COVID-19 pandemic. Moreover, there seems to be a positive outlook for the pharmaceutical drug pipeline in the next 10 years.

Reflection

Our investment in VHT reflected our team's willingness to look past near-term macroeconomic trends and political uncertainties and instead focus on the long-term outlook for healthcare in the next ten years. Despite the uncertainties that healthcare will likely face with the new Trump administration, we increased our allocation into VHT due to the confidence in the sector's strong pipeline. This trade shaped our final strategy by emphasizing the importance of a horizon-based approach that prioritizes long term growth even when current trends appear unpromising.

Stock

Sony Group Corp (SONY)

Trade Note

Our DCF model calculates the fair value of SONY to be $28.37 a share, a large difference from the current price of around $20. Furthermore, the recovery of the Japanese economy coupled with investments in diversified avenues such as SONY Music and PlayStation provide a positive outlook for SONY.

Reflection

Our initial approach to forecasting, including SONY and other companies focused heavily on future growth plans. However, a Morningstar article illustrating how SONY is building their economic MOAT through music and games revealed that competitive advantage is critical for accurate forecasting. MOAT is not just about a company's ability to sustain itself, but also its capacity to dominate and capture opportunities in its industry. This insight led us to incorporate MOAT into our confidence scores for each stock, which will be shown later in the report.

Stock

GE Vernova Inc (GEV)

Trade Note

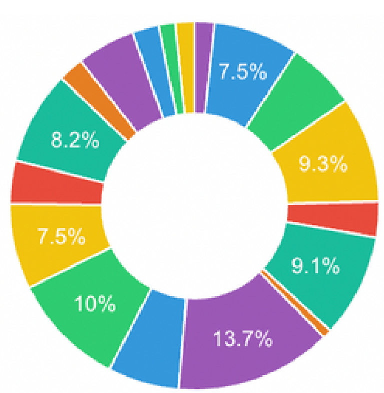
Our DCF model estimates the fair value of GEV to be approximately $360 compared to its current price of $329. Our fair value price is largely attributed to its strong value proposition, as both electrical utility companies and magnificent 7 stocks are making significant investments in electricity.

Reflection

Our investment in GEV was driven by our understanding that electricity demand will surge due to AI infrastructure development. In fact, JP Morgan's "Guide to the Market" from 2024 projects that electricity demand, driven in part by AI growth, will rise by 30% from 2022 to 2026. Initially, we focused on electrical utility companies, but digging further we realized the potential of energy equipment companies like GEV. This trade taught us that when trying to find outside stocks with macroeconomic data, it's essential to research the entire supply chain for investment opportunities.

WInS Portfolio

Sector	Ticker	Value
Technology	MSFT	13.73
Technology	SONY	9.31
Technology	ASML	5.12
Consumer Cyclical	ABNB	1.69
Consumer Cyclical	AMZN	2.34
Financials	PYPL	9.09
Financials	AFL	1.44
Utilities	GEV	7.51
Healthcare	PFE	0.8
Healthcare	IQV	10.03
Health ETF	VHT	6.06
Information Technology	CRM	8.21
Consumer Defensive	LRN	6.27
Materials	FMC	3.79
Small value ETF	AVUV	2.24
Large blend ETF	VTI	7.49
Large value ETF	VWO	1.77
Large value ETF	SCHD	3.12
Left over Cash	**$6,106.06**	**6.1% of 100,000**



RECOMMENDED PORTFOLIO

Sector	Ticker	Value
Technology	MSFT	17.17
Financials	PYPL	12.23
Technology	SONY	10.70
Health Care	IQV	9.80
Materials	FMC	8.12
Energy	GEV	8.00
Consumer Discretionary	ABNB	5.52
Consumer Discretionary	LRN	3.42
Technology	CRM	5.04
Broad Market ETF	VTI	7.00
Health Care ETF	VHT	6.00
Small Cap Value ETF	AVUV	3.10
Large Cap Value ETF	SCHD	2.20
Broad Market ETF	VWO	1.70

Why is it Different from our WInS portfolio

Our WInS portfolio initially reflected an exploratory phase where we focused on undervalued companies based on our model without deeply considering client fit. Earlier investments, such as Amazon and Pfizer, conflicted with our client's ethical priorities. Amazon's establishment of headquarters on native land and Pfizer's history of unethical experimentation in Africa highlighted misalignments with Ladi Ayoola's values. As we refined our strategy, we found it increasingly difficult to identify strong opportunities in the market that fit our client's views. This led us to a shorter recommended portfolio. However, our ETF trades remained consistent, as we developed the ETF strategy in the latter half of the investment period.

Intro

Our philosophy of stock investing coming into this competition is that an intrinsic value, or fair value estimate, is equal to the value of the cash a business can generate across its lifespan. While this remains our guiding principle, the case study highlighted the importance of also considering shorter-term growth opportunities. In addition to identifying undervalued companies with strong potential over the next 10–15 years, we are focusing on companies that can deliver substantial growth within the next five years to align with Ladi Ayoola's phase 1 goals. Below is our overall strategy to do just that.

Screener

The majority of our strategy comes from our fair value model that is described below, which requires extensive computing for the machine learning model, along with an in-depth review of 10-K filings, 5-K filings, articles, earnings calls, and other relevant material. Instead of doing extensive research for all 500 stocks on the provided list, we applied filters to narrow down the companies with the goal of finding companies providing consistent growth and financial stability.

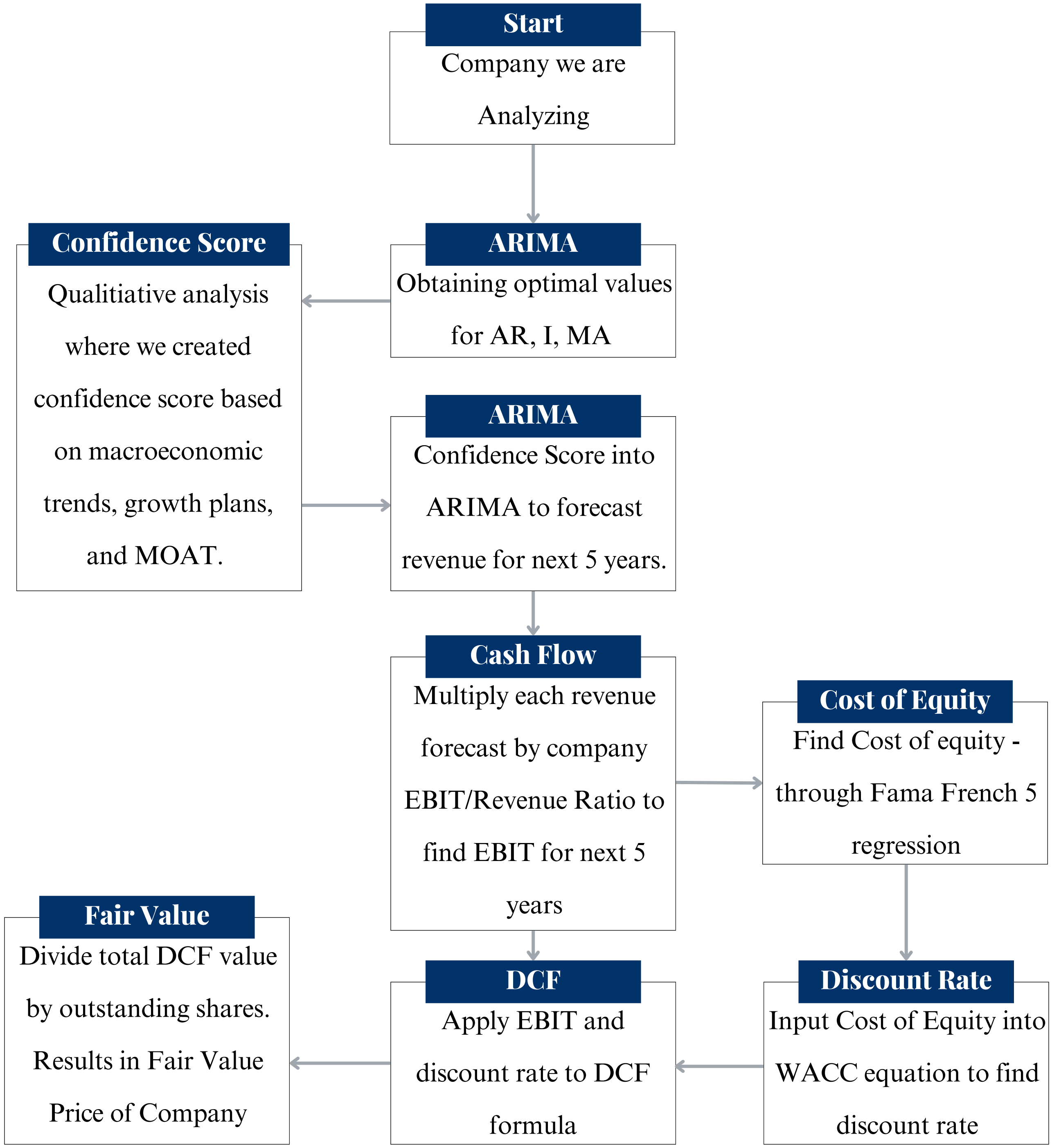
Below is the filter the companies had to pass to be further analyzed in the DCF stage.

Net Profit Margin	Return on Investment	Sales Growth	Forward P/E	Market Cap
Over 5%	Over 5%	over the past 5 years being positive (>0%)	Under 40	+Mid (over $2 billion)

This left us with 139 companies. Initially, we included a market cap filter that only included large cap companies(+10 billion market cap). However, after learning more about the high valuation of the U.S. market with the SPY reaching all time highs in P/E ratios, we identified that there may be value in smaller sized companies.

Fair Value Model Flow Chart:

FAIR VALUE FLOW CHART



CASH FLOW FORECASTS

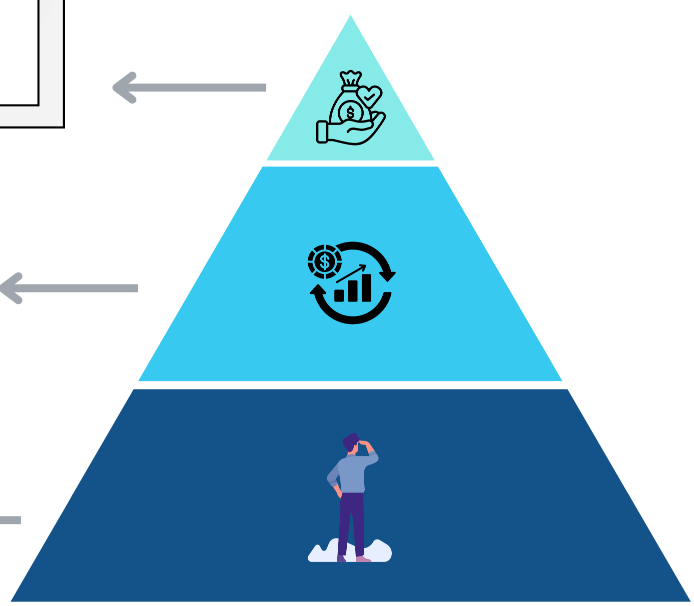
Revenue Forecasting

AutoRegressive Integrated Moving Average (ARIMA) is a statistical tool we utilized to streamline revenue forecasting, offering a faster alternative to the traditional, labor-intensive process of projecting every component of a company's financials in a typical DCF model. While ARIMA excels at identifying and leveraging historical patterns to predict future revenues, it inherently lacks a forward-looking perspective that accounts for qualitative factors such as industry, macroeconomic trends, growth potential, etc. To address this limitation, we incorporated a confidence score (ranging from 1 to 100) into the ARIMA model which enhances the model's alignment with real-world growth scenarios, as the adjusted fair value price reflects our qualitative judgment alongside the quantitative output. The process for determining this score is detailed in the accompanying graphic. After these forecasts for the next 5 years are completed, we need to find the free cash flow for that year. To do this, we calculated EBIT by multiplying the revenue by the EBIT/Revenue Ratio. Then we subtracted taxes, added back estimated non-cash expenses like depreciation and amortization, and adjusted for changes in working capital.

(20%) : Porter's five forces was used to assess the businesses' current MOAT, and how they plan on building their MOAT to become a more dominant player in the industry.

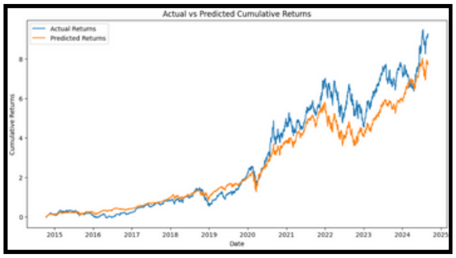
(30%) : PESTLE Analysis was used in evaluating macroeconomic trends, and whether a company is in the right position to capitalize on these opportunities.

(50%) : Looked at 10k Fillings, 5k Fillings, Earning Calls, and articles to evaluate future plans and growth opportunities.



Discount Rate

The discount rate in the DCF model is the Weight Average Cost of Capital(WACC). The most important part of WACC is the cost of equity calculation, which is the return required for an investor based on the risk associated with the stock. This is typically done with CAPM, but for our model, we used the Fama French 5 Factor Model, as we believe it's a better measure of risk, as it finds Betas for five different factors rather than just one.



We employ the Fama-French 5-Factor model to compute daily cost of equity based on the stock's excess returns over the risk-free rate and its sensitivity to multiple factors, including market risk, size, value, etc.

WACC Equation

After cost of equity is calculated, we input it into the WACC equation that includes various parts, such as risk-free rate, tax rate, cost of debt, ete. This returns the discount rate for the DCF model.

Fair Value Price

Using the five year cash flow forecasts and the discount rate, we applied it to the Discounted Cash Flow Formula. wE assumed a growth rate of 4%.

DCF

Since DCF is our valuation of the company, we divide that by total outstanding shares to get the fair value price of the company. If the company's current price/fair value ratio is less than one, then it moves on to the client fit analysis stage.

$$\text{DCF} = \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \cdots + \frac{CF_n}{(1+r)^n} + \left(\frac{CF_n \times (1+g)}{r-g}\right)/(1+r)^n$$

The Discounted Cash Flow Formula

CLIENT FIT

Why FRIES?

After putting our stocks through both a qualitative and quantitative analysis, we wanted to create a holistic way to analyze the client fit of each stock. To this end, we decided to turn our name, “The Fama French Fries”, into an acronym that would highlight the qualities in each company that would align with Ladi Ayoola’s principles. Each section is rated on a scale of 1 to 5, with 1 indicating minimal or no alignment in that area, and 5 representing exceptional performance that exceeds expectations.

F

Faultless Standards: Evaluates the companies commitment to ethical practices and sustainability. They should not only comply with legal standards, but go above and beyond with inititatives that enhance transparency and create fair labor practices.

R

Relevance to Client and Wife Interests: Evaluates the companies relevance to Ladi Ayoola and Mosun Ayoola’s interests which include: Electrical Engineering, Entrepreneurship, and Fashion. The sector along with the the companies positive impact in the field is evaluated.

I

Infrastructure Support in Africa: Evaluates the companies contributions to improving infrastructure in Africa with an emphasis on transportation networks and energy grids which are areas needing support in Lugbe, Abuja, Nigeria.

E

Educational Support in Africa: Evaluates the companies focus on improving educational access and quality in Africa. Strong emphasis on companies that put a strong emphasis on empowering the African youth through educational intiatives.

S

Social Impact Globally: Evaluates the companies broader social impacts and global philanthropic efforts. Emphasis on corporate social responsibility (CSR) programs that address large-scale issues like poverty and inequality.

Client Fit Cont.

It's unrealistic for companies to excel in all these categories; however, based on the client's priorities, they must demonstrate strong ethical standards (F) and a commitment to social contributions (S). These two aspects are critical. For the remaining three categories—client interests (R), education (E), and infrastructure (I)—the company must either excel in at least one area or achieve a score of 3/5 or higher in at least two of these categories.

Finding Outside Stocks

Our approach to identifying stocks outside the provided list aligned investments with Ladi Ayoola's core values: entrepreneurship, education, infrastructure, and electrical engineering. To ensure these selections were at the same level as our previous stocks, we conducted a DCF analysis for each selection.

The first stock we identified outside the list was Airbnb. Airbnb fosters initiatives like the Airbnb Entrepreneurship Academy, which not only empowers local entrepreneurs with skills they need to succeed, but also drives economic growth in underrepresented areas. The second stock we identified was Stride, a company that provides innovative online education solutions. Stride is committed to making education accessible to anyone, offering a range of online K-12 schools, career training, and adult learning programs. The last stock we identified was General Electric Vernova (GEV). GEV exemplifies electrical engineering through its advanced technologies, such as the GE H-Class gas turbines, which improve power generation efficiency while reducing emissions. The company's infrastructure efforts are particularly evident in Africa, where GEV is working with countries like Kenya and Egypt to enhance energy access.

Selecting ETFs Using The Fama French 5 Factor Model

The Fama French 5 factor model is widely used to identify factor premiums across entire portfolios, making it particularly well-suited for analyzing ETFs, which are essentially collections of various stocks. Using this model, we evaluated ETFs on the provided list and drew insights about various markets and industries. From research, we learned that emerging markets tend to have strong premiums, broad-market ETFs perform better than large-cap ETFs, and small-cap value ETFs often show great premiums. Our findings lined up with these conclusions as AVUV, a small-cap value ETF, showed the best premiums in our analysis, as shown in the table below.

Based on this, we invested in a mix of ETFs, including a broad market ETF, an emerging market ETF, and others with strong factor premiums. With the U.S. market being extremely expensive right now—especially with large-cap growth stocks—this is the perfect time to apply the Fama French 5 Factor Model. By focusing on ETFs with favorable premiums, we're positioned to take advantage of better-priced opportunities in other segments and markets.

Annualized Alpha	MKT-RF	SMB	HML	RMW	CMA
0.43%	1.03	0.89	0.55	0.34	-0.04

Portfolio Allocation

Ladi Ayoola's financial strategy is centered on achieving long-term growth while accommodating mid-term returns to support his plans for starting a family. By conducting an in-depth risk profile for Ladi Ayoola, we decided on an 80/20 allocation of individual stocks vs. ETFs, balancing a robust growth potential with healthy diversification. By maximizing the Sharpe ratio along the efficient frontier, our method calculated the optimal weighting for each stock to achieve maximum returns while taking the least amount of risk. Then, we used historical returns of the ETFs with the risk premiums derived from the Fama French 5-factor model to determine the optimal ETF allocation, ensuring diversification across multiple markets and sectors while aligning with Ladi Ayoola's growth strategy and risk tolerance.

Throughout this proposal we have laid out our plans to invest in high-quality undervalued companies that have the potential to generate significant returns However, the returns that we generate are only as important as the impact that the money creates. That's why we have created a comprehensive plan for Phase 3 in Lugbe, Abuja, Nigeria. In tandem with Ladi Ayoola's love for electrical engineering and fashion, we plan to add to the multi-hub center by creating two buildings that will serve as a place for the Nigerian youth to get work experience while contributing to the culture and economy of the local region.

Building 1

The first building will serve as the electrical center of the community where Nigerian youth will have the opportunity to learn how to design, install, and maintain solar panels, inverters, and battery systems. This will allow them to get hands-on experience while improving the electrical reliability of the area.

Building 2

The second building will serve as a display of cultural heritage, as the Nigerian youth will learn about making traditional Nigerian clothing and will apply the knowledge they gained to make several short films about the area.

By conducting a thorough analysis of Ladi Ayoola's, we have discovered that he enjoys soccer. Therefore, to share his love for the sport and add a vibrancy of the region, a soccer field will be constructed to allow the community to get together and have a good time. Finally, we propose the addition of an incubator program that partners the youth interested in electrical engineering with those interested in filmmaking and fashion to create innovative products. Partnering with Plugtent, this incubator will invest in the Nigerian youth and create a way for them to monetize their ideas.

WORKS CITED

Ahern, D. (2024, September 24). How to calculate market value of debt (with real-life examples). Investing for Beginners 101. https://einvestingforbeginners.com/market-value-of-debt-daah/

Alex, T. (2022, September 16). A brief comparison of the Nigerian and United States electric power sectors. Medium. https://energypoliticking.com/a-brief-comparison-of-the-nigerian-and-united-states-electric-power-sectors-ccb7ff1f6586

Bloom, D. (2024, August 27). Sony builds moat with music and games as Morningstar raises fair value. Forbes. https://www.forbes.com/sites/dbloom/2024/07/05/sony-builds-moat-with-music-games-as-morningstar-raises-fair-value/

Guide to the markets. Welcome. (n.d.). https://am.jpmorgan.com/us/en/asset-management/adv/insights/market-insights/guide-to-the-markets/

Hargrave, M. (n.d.). Weighted average cost of capital (WACC) explained with formula and example. Investopedia. https://www.investopedia.com/terms/w/wacc.asp

Market perspectives. Our investment and economic outlook, December 2024. (n.d.). https://advisors.vanguard.com/insights/article/series/market-perspectives

Owanbe Community. (2018, March 27). This traditionally styled shoot will make you fall in love with the richness of Nigerian culture #TheAkugbe. https://owanbecommunity.org/this-traditionally-styled-shoot-will-make-you-fall-in-love-with-the-richness-of-nigerian-culture-theakugbe/

Search filings. U.S. Securities and Exchange Commission. (n.d.). https://www.sec.gov/search-filings